

Практикум

B4:archive

Every past-paper question on this topic, grouped by technique. Not a practicum — a drill. There is no theory here and no ladder to climb: the theory is in Practicum B4, and this notebook is what you open afterwards, when the only thing left is to do them all until the moves are automatic.

О практикуме

Обе темы взяты целиком: 147 баллов, 62 блока. По отдельности ни одна не дотягивает до нижней границы — 113 и 34, — а вместе они и в архиве стоят одним вопросом: ноябрь 2021 Paper 2 Q10 просит пересечения, потом вертикальную асимптоту, потом наклонную, потом эскиз, и это 11 баллов из 18.

Заданий

0

Раздел решений

включён

Интерактивная версия

kaggle.com/code/artemkonukhov/ib-math-praktikum-b4-curve-sketching

Собрано из ноутбука репозитория `ib-math-aa-hl-past-papers`. Проверки в PDF не работают — они для интерактивной версии.

B4 archive: curve sketching and asymptotes

Every past-paper question on this topic, grouped by technique. Not a practicum — a drill. There is no theory here and no ladder to climb: the theory is in *Practicum B4*, and this notebook is what you open afterwards, when the only thing left is to do them all until the moves are automatic.

What is inside. The whole of `functions.curve_sketching` and the whole of `functions.asymptotes` from the AA HL archive, sessions May 2021 — November 2025: **54 questions, 127 marks**, in nine sections, one section per technique.

The corpus records 62 blocks and 147 marks. The difference is the November 2023 session, which sits in the archive as two zonal copies of one paper; the duplicates are merged here, and what is left is what was actually asked. Where a

question appears in both copies the source line says so.

How to work. Read the question, answer in the cell below it, run the cell. The check is not a comparison with a stored answer — it goes back to the function and asks it. An asymptote is right when the curve actually approaches it, a range is right when every value in it is attained and none outside it is, a sketch is right when the features you listed are the features the function has. So an answer arrived at by a wrong route still has to be true.

Leave a cell blank and it prints  and moves on, which means you can run the whole notebook top to bottom on the first open and nothing breaks.

The solutions are at the end, numbered the same way. Open one after you have worked the question, not before — and read the markscheme note in it, because that is where the marks actually are.

The nine sections.

§	technique	questions	marks
1	Read the vertical and horizontal asymptotes off a rational function	8	12
2	Find an asymptote that cannot be read off	2	3
3	Find the oblique asymptote	4	13
4	Sketch a rational curve	10	26
5	Find the range	4	13
6	Sketch a curve from its formula, labelling what was asked for	11	33
7	Sketch using the second derivative	1	3
8	Sketch a curve that is not a function	5	10
9	Count the roots of a family	9	14

One sentence to carry into the exam: **the marks are for the list, not for the curve.** Intercepts, turning points, asymptotes — write that list down before drawing anything.

```

import sys
sys.path.append('.')          # from
practicum/functions to practicum/kit/
import sympy as sp            # the escape
hatch: anything not in kit is in sp
from kit import *              # checks +
Interval, Union, FiniteSet, Eq, solveset

import matplotlib.pyplot as plt
import numpy as np

language('en')                # this
notebook is in English, and so are the checks

a, b, c, d, m, t = symbols('a b c d m t')
A = Symbol('A', positive=True)

# Draw the true curve so you can compare your
# sketch with it. Pass
# expressions, or (expression, label) pairs;
# `lines` takes asymptotes as
# ('v', 3) or ('h', -2) or ('o', x/2 - 1);
# `marks` takes points to dot.
def show(*curves, span=(-6, 6), ylim=None,
size=(6, 3.4), marks=(), lines=()):
    grid = np.linspace(float(span[0]),
float(span[1]), 2000)
    fig, ax = plt.subplots(figsize=size)
    for item in curves:
        expr, name = item if isinstance(item,
tuple) else (item, None)
        g = lambdify(x, expr, 'math')
        vals = []
        for u in grid:
            try:

```

```

        v = float(g(float(u)))
    except (ValueError, TypeError,
ZeroDivisionError, OverflowError):
        v = float('nan')
    vals.append(v if abs(v) < 1e6
else float('nan'))
    ax.plot(grid, vals, lw=1.7,
label=name or f'y = {expr}')
    for kind, where in lines:
        if kind == 'v':
            ax.axvline(float(where),
color='0.55', ls='--', lw=1)
        elif kind == 'h':
            ax.axhline(float(where),
color='0.55', ls='--', lw=1)
        else:
            g = lambdify(x, where, 'math')
            ax.plot(grid, [g(float(u)) for u
in grid], color='0.55', ls='--', lw=1)
            for point in marks:
                ax.plot(float(point[0]),
float(point[1]), 'o', ms=5, color='0.2')
                ax.axhline(0, color='0.6', lw=.8)
                ax.axvline(0, color='0.6', lw=.8)
                ax.grid(alpha=.3)
            if ylim:
                ax.set_ylim(float(ylim[0]),
float(ylim[1]))
            ax.legend(fontsize=8)
            plt.tight_layout()
            plt.show()

print('ready; sympy', sp.__version__)
print('an asymptote:          ', Eq(y, x/2 -
Rational(17, 2)))
print('a range:              ',

```

```
Union(Interval(-oo, -1), Interval(2, oo)))  
print('a sketch as features:',  
{ 'x_intercepts': [2, 12], 'minima': [(5.66,  
-1.34)] })  
print('sketch keys:', 'x_intercepts',  
y_intercept, maxima, minima, cusps',  
      'endpoints, vertical_asymptotes',  
horizontal_asymptotes, oblique_asymptotes')
```

1 • Read the vertical and horizontal asymptotes off a rational function

"Write down the equation of the vertical asymptote." Set the denominator to zero; compare the degrees for the horizontal one. **8 questions, 12 marks, and eight of them are Paper 1.**

Every answer here is an equation. A number scores zero — the markscheme says *"must be written as an equation"* in as many words.

1.1 · November 2021, Paper 1, Q2(a)
· 2 marks · no calculator

The function f is defined by $f(x) = \frac{2x + 4}{3 - x}$,
where $x \in \mathbb{R}$, $x \neq 3$.

Write down the equation of

- (i)** the vertical asymptote of the graph of f ;
- (ii)** the horizontal asymptote of the graph of f .

```
q1_1 = [...] # both
asymptotes, as equations

verify_asymptotes('1.1', q1_1, (2*x + 4)/(3 -
x))
```

1.2 · November 2021, Paper 2,
Q10(b) · 1 mark · calculator

Consider the function $f(x) = \frac{x^2 - x - 12}{2x - 15}$,
where $x \in \mathbb{R}$, $x \neq \frac{15}{2}$.

Write down the equation of the vertical
asymptote of the graph of f .

```
q1_2 = ... # the vertical
asymptote, as an equation

verify_asymptotes('1.2', q1_2, (x**2 - x -
12)/(2*x - 15), kinds=('vertical',))
```

1.3 · May 2022 TZ2, Paper 1, Q3(a) · 2 marks · no calculator

A function f is defined by $f(x) = \frac{2x - 1}{x + 1}$, where $x \in \mathbb{R}$, $x \neq -1$.

The graph of $y = f(x)$ has a vertical asymptote and a horizontal asymptote. Write down the equation of

(i) the vertical asymptote; **(ii)** the horizontal asymptote.

```
q1_3 = [...]

verify_asymptotes('1.3', q1_3, (2*x - 1)/(x +
1))
```

1.4 · May 2023 TZ1, Paper 1, Q1(b) · 2 marks · no calculator

The function f is defined by $f(x) = \frac{7x + 7}{2x - 4}$ for $x \in \mathbb{R}$, $x \neq 2$.

For the graph of $y = f(x)$, write down the equation of

(i) the vertical asymptote; **(ii)** the horizontal asymptote.

```
q1_4 = [...]
```

```
verify_asymptotes('1.4', q1_4, (7*x + 7)/(2*x  
- 4))
```

**1.5 · May 2023 TZ2, Paper 1, Q2(a) ·
2 marks · no calculator**

A function f is defined by $f(x) = 1 - \frac{1}{x - 2}$,
where $x \in \mathbb{R}$, $x \neq 2$.

The graph of $y = f(x)$ has a vertical asymptote and a horizontal asymptote.
Write down the equation of

(i) the vertical asymptote; **(ii)** the horizontal asymptote.

```
q1_5 = [...]
```

```
verify_asymptotes('1.5', q1_5, 1 - 1/(x - 2))
```

**1.6 · November 2023, Paper 2,
Q11(a) · 1 mark · calculator**

Recorded twice in the corpus, once under TZ1 and once under TZ2. It is one paper.

Consider the function defined by $f(x) = \frac{x^2 - 14x + 24}{2x + 6}$, where $x \in \mathbb{R}$, $x \neq -3$.

State the equation of the vertical asymptote on the graph of f .

```
q1_6 = ...
```

```
verify_asymptotes('1.6', q1_6, (x**2 - 14*x +  
24)/(2*x + 6), kinds=('vertical',))
```

**1.7 · May 2024 TZ2, Paper 1, Q5(a) ·
1 mark · no calculator**

A function f is defined by $f(x) = \frac{2(x+3)}{3(x+2)}$, where $x \in \mathbb{R}$, $x \neq -2$. The graph of $y = f(x)$ is given in the paper.

Write down the equation of the horizontal asymptote.

```
q1_7 = ...
```

```
verify_asymptotes('1.7', q1_7, 2*(x + 3)/(3*(x + 2)), kinds=('horizontal',))
```

1.8 · May 2025 TZ2, Paper 1, Q1(b) · 1 mark · no calculator

The function f is defined by $f(x) = \frac{3x - 2}{2x + 1}$ for $x \in \mathbb{R}$, $x \neq -\frac{1}{2}$.

Write down the equation of the horizontal asymptote.

```
q1_8 = ...
```

```
verify_asymptotes('1.8', q1_8, (3*x - 2)/(2*x + 1), kinds=('horizontal',))
```

2 · Find an asymptote that cannot be read off

No denominator to set to zero, no degrees to compare — the answer is a limit and nothing else. **2 questions, 3 marks.**

2.1 · May 2021 TZ2, Paper 2, Q12(b) · 2 marks · calculator

A function f is defined by $f(x) = \arcsin\left(\frac{x^2 - 1}{x^2 + 1}\right)$, $x \in \mathbb{R}$.

By considering limits, show that the graph of $y = f(x)$ has a horizontal asymptote and state its equation.

```
q2_1 = ... # the  
horizontal asymptote, as an equation
```

```
verify_asymptotes('2.1', q2_1, asin((x**2 -  
1)/(x**2 + 1)), kinds=('horizontal',))
```

2.2 · November 2023, Paper 1, Q10(a) · 1 mark · no calculator

Recorded twice in the corpus, TZ1 and TZ2. One paper.

The functions f and g are defined by

$$f(x) = \ln(2x - 9),$$

$$x > \frac{9}{2}$$

$$g(x) = 2 \ln x - \ln d,$$

$$x > 0,$$

$$d \in \mathbb{R}^+.$$

State the equation of the vertical asymptote to the graph of $y = g(x)$.

```
q2_2 = ...
```

```
dpos = Symbol('d', positive=True)
verify_asymptotes('2.2', q2_2, 2*log(x) -
log(dpos), kinds=('vertical',))
```

3 · Find the oblique asymptote

The numerator's degree is exactly one more than the denominator's. Divide: the quotient is the line, the remainder is what vanishes. **4 questions, 13 marks — and all of them are on calculator papers, where the calculator can do nothing at all.**

3.1 · May 2021 TZ1, Paper 2, Q11(e) · 4 marks · calculator

The function g is defined by $g(x) = \frac{4x^2 - 1}{3x + 2}$, for $x \in \mathbb{R}$, $x \neq -\frac{2}{3}$.

Find the equations of **all** the asymptotes on the graph of $y = g(x)$.

```
q3_1 = [...] # all of them
- and there may be fewer kinds than you
expect

verify_asymptotes('3.1', q3_1, (4*x**2 -
1)/(3*x + 2))
```

3.2 · November 2021, Paper 2, Q10(c) · 4 marks · calculator

Consider $f(x) = \frac{x^2 - x - 12}{2x - 15}$, $x \neq \frac{15}{2}$.

The oblique asymptote of the graph of f can be written as $y = ax + b$ where $a, b \in \mathbb{Q}$. Find the value of a and the value of b .

```
q3_2 = ... # the whole
line, as an equation

verify_asymptotes('3.2', q3_2, (x**2 - x -
12)/(2*x - 15), kinds=('oblique',))
```

3.3 · November 2023, Paper 2, Q11(c) · 4 marks · calculator

Recorded twice in the corpus, TZ1 and TZ2. One paper.

Consider $f(x) = \frac{x^2 - 14x + 24}{2x + 6}$, $x \neq -3$.

The graph of f has an oblique asymptote of the form $y = ax + b$, where $a, b \in \mathbb{Q}$. Find

the value of a and the value of b .

```
q3_3 = ...
```

```
verify_asymptotes('3.3', q3_3, (x**2 - 14*x +  
24)/(2*x + 6), kinds=('oblique',))
```

3.4 · May 2025 TZ2, Paper 3, Q1(d) (ii) · 1 mark · calculator

Consider the curve $y = \frac{x(x^2 - A)}{x^2 + A}$, where A is a positive constant and $x \in \mathbb{R}$.

Part (d)(i) has just established the identity $x - \frac{2Ax}{x^2 + A} \equiv \frac{x(x^2 - A)}{x^2 + A}$.

Hence, determine the equation of the oblique asymptote to the curve.

```
q3_4 = ...
```

```
# one answer
```

```
for every A at once
```

```
verify_asymptotes('3.4', q3_4, x*(x**2 -  
A)/(x**2 + A), kinds=('oblique',))
```

4 · Sketch a rational curve

Asymptotes first, dashed and labelled;
then the intercepts; then the turning

points; then join the branches, one region at a time. That order is not advice — it is the order the markscheme pays in. **10 questions, 26 marks.**

The check reads your sketch as the list of features you claim, and it catches both kinds of error separately: a feature named that is not there, and a feature there that you did not name. What it cannot see is the shape *between* the features, so every task here prints the true curve underneath. Compare.

4.1 · May 2021 TZ1, Paper 2, Q11(d) · 5 marks · calculator

The function f is defined by $f(x) = \frac{3x + 2}{4x^2 - 1}$, for $x \in \mathbb{R}$, $x \neq p$, $x \neq q$.

Sketch the graph of $y = f(x)$ for $-3 \leq x \leq 3$, showing the values of any axes intercepts, the coordinates of any local maxima and local minima, and giving the equations of any asymptotes.

```

q4_1 = {...}      # keys: vertical_asymptotes,
horizontal_asymptotes, x_intercepts,
                    # y_intercept, maxima,
minima

f41 = (3*x + 2)/(4*x**2 - 1)
verify_sketch('4.1', q4_1, f41)

show(f41, span=(-3, 3), ylim=(-6, 6), size=
(6, 3.6),
     lines=[('v', Rational(-1, 2)), ('v',
Rational(1, 2)), ('h', 0)])

```

4.2 · November 2021, Paper 1, Q2(c)

· 1 mark · no calculator

Sketch the graph of $f(x) = \frac{2x + 4}{3 - x}$ on the axes provided (both axes run from -15 to 15).

The question asks only for the shape; list the features you would put on the sketch and let the check confirm them.

```

q4_2 = {...}      # keys: vertical_asymptotes,
horizontal_asymptotes,
                    # x_intercepts, y_intercept

f42 = (2*x + 4)/(3 - x)
verify_sketch('4.2', q4_2, f42)

show(f42, span=(-15, 15), ylim=(-15, 15),
     lines=[('v', 3), ('h', -2)])

```

4.3 · November 2021, Paper 2, Q10(d) · 3 marks · calculator

Sketch the graph of $f(x) = \frac{x^2 - x - 12}{2x - 15}$ for $-30 \leq x \leq 30$, clearly indicating the points of intersection with each axis and any asymptotes.

```
q4_3 = {...}      # keys: vertical_asymptotes,
oblique_asymptotes,
                  # x_intercepts, y_intercept

f43 = (x**2 - x - 12)/(2*x - 15)
verify_sketch('4.3', q4_3, f43)

show(f43, span=(-30, 30), ylim=(-30, 30),
      lines=[('v', Rational(15, 2)), ('o', x/2
+ Rational(13, 4))])
```

4.4 · May 2022 TZ2, Paper 1, Q3(b) · 3 marks · no calculator

On the set of axes provided (-6 to 6 both ways), sketch the graph of $y = f(x)$ where $f(x) = \frac{2x - 1}{x + 1}$.

On your sketch, clearly indicate the asymptotes and the position of any points of intersection with the axes.

```

q4_4 = {...}      # keys: vertical_asymptotes,
horizontal_asymptotes,
                  # x_intercepts, y_intercept

f44 = (2*x - 1)/(x + 1)
verify_sketch('4.4', q4_4, f44)

show(f44, span=(-6, 6), ylim=(-6, 6), lines=
[('v', -1), ('h', 2)])

```

4.5 · May 2022 TZ2, Paper 1, Q11(a) · 6 marks · no calculator

A function f is defined by $f(x) = \frac{1}{x^2 - 2x - 3}$, where $x \in \mathbb{R}$, $x \neq -1$, $x \neq 3$.

Sketch the curve $y = f(x)$, clearly indicating any asymptotes with their equations. State the coordinates of any local maximum or minimum points and any points of intersection with the coordinate axes.

```

q4_5 = {...}      # keys: vertical_asymptotes,
horizontal_asymptotes,
                  # x_intercepts, y_intercept,
maxima

f45 = 1/(x**2 - 2*x - 3)
verify_sketch('4.5', q4_5, f45)

show(f45, span=(-4, 6), ylim=(-3, 2), lines=
[('v', -1), ('v', 3), ('h', 0)])

```

4.6 · May 2023 TZ2, Paper 1, Q2(c) · 1 mark · no calculator

On the set of axes provided, sketch the graph of $y = f(x)$ where $f(x) = 1 - \frac{1}{x-2}$, showing all the features found in parts (a) and (b) — the two asymptotes and the two intercepts.

```

q4_6 = {...}      # keys: vertical_asymptotes,
horizontal_asymptotes,
                  # x_intercepts, y_intercept

f46 = 1 - 1/(x - 2)
verify_sketch('4.6', q4_6, f46)

show(f46, span=(-5, 9), ylim=(-5, 6), lines=
[('v', 2), ('h', 1)])

```

**4.7 · November 2023, Paper 2,
Q11(d) · 4 marks · calculator**

Recorded twice in the corpus, TZ1 and TZ2. One paper.

Sketch the graph of $f(x) = \frac{x^2 - 14x + 24}{2x + 6}$ for $-50 \leq x \leq 50$, showing clearly the asymptotes and any intersections with the axes.

```
q4_7 = {...}      # keys: vertical_asymptotes,
oblique_asymptotes,
                  # x_intercepts, y_intercept

f47 = (x**2 - 14*x + 24)/(2*x + 6)
verify_sketch('4.7', q4_7, f47)

show(f47, span=(-50, 50), ylim=(-60, 40),
      lines=[('v', -3), ('o', x/2 -
Rational(17, 2))])
```

**4.8 · May 2025 TZ2, Paper 3, Q1(a)(i)
(iii) · 1 + 2 marks · calculator**

Consider the curve given by $y = \frac{x(x^2 - 16)}{x^2 + 16}$.

(i) Sketch the curve of y for $-10 \leq x \leq 10$.

(iii) State the coordinates of the local maximum point and the coordinates of

the local minimum point.

```
q4_8 = {...}      # keys: x_intercepts,
y_intercept, maxima, minima

f48 = x*(x**2 - 16)/(x**2 + 16)
verify_sketch('4.8', q4_8, f48)

show(f48, span=(-10, 10), ylim=(-10, 10),
lines=[('o', x)])
```

4.9 · May 2025 TZ2, Paper 3, Q1(a) (ii) · 1 mark · calculator

State the coordinates of the points where the same curve $y = \frac{x(x^2 - 16)}{x^2 + 16}$ crosses the x -axis.

```
q4_9 = [...]      # the x-
coordinates

verify_sketch('4.9', {'x_intercepts': q4_9},
x*(x**2 - 16)/(x**2 + 16))
```

4.10 · November 2025 TZ1, Paper 3, Q1(d)(i) · 1 mark · calculator

The curve H has equation $y = \frac{1}{x}$, where $x \in \mathbb{R}$, $x \neq 0$.

Sketch the curve H .


```
q4_10 = {...}      # keys: vertical_asymptotes,
horizontal_asymptotes

verify_sketch('4.10', q4_10, 1/x)

show(1/x, span=(-6, 6), ylim=(-6, 6), lines=
[('v', 0), ('h', 0)])
```

5 • Find the range

The same picture read sideways: off the vertical axis instead of the horizontal one. **4 questions, 13 marks — and the mark is usually one symbol.** A turning point is attained and takes $\leq$; a horizontal asymptote is not and takes $<$.

The check tests every endpoint of your set three times over — the endpoint itself and a step to either side of it — because that is the only place this topic is ever wrong.

5.1 • May 2021 TZ2, Paper 2, Q4(a) • 2 marks • calculator

The functions f and g are defined for $x \in \mathbb{R}$ by $f(x) = 6x^2 - 12x + 1$ and $g(x) = -x + c$,

where $c \in \mathbb{R}$.

Find the range of f .

```
q5_1 = ... # a set or an
inequality: Interval(...), (y >= ...)

verify_range('5.1', q5_1, 6*x**2 - 12*x + 1)
```

5.2 · May 2023 TZ2, Paper 2, Q8(a) · 4 marks · calculator

A function g is defined by $g(x) = \frac{2x - 5}{x^2 - 3}$,
where $x \in \mathbb{R}$, $x \neq \pm\sqrt{3}$.

Determine the range of g .

```
q5_2 = ... # exact form;
the calculator gives 3 s.f., the algebra
gives surds

verify_range('5.2', q5_2, (2*x - 5)/(x**2 -
3))
```

5.3 · November 2023, Paper 2, Q11(e) · 4 marks · calculator

The corpus records this part only under TZ2, but it is printed in both copies of the paper.

Find the range of f , where $f(x) = \frac{x^2 - 14x + 24}{2x + 6}$, $x \neq -3$.

```
q5_3 = ...
```

```
verify_range('5.3', q5_3, (x**2 - 14*x + 24)/(2*x + 6))
```

5.4 · November 2025 TZ3, Paper 1, Q9(b) · 3 marks · no calculator

Consider the function defined by $f(x) = \frac{1}{2}x^2 + kx + 13$, where $x \in \mathbb{R}$ and $k \in \mathbb{Z}^+$. Part (a) has shown that if $f(x) = 0$ has no real roots then the greatest possible value of k is 5. Take $k = 5$.

(i) Write down the equation of the axis of symmetry of the graph of f .

(ii) Hence, or otherwise, determine the coordinates of the minimum point on the graph of f .

```
q5_4a = ... # the axis of symmetry, as an equation
q5_4b = {...} # key: minima

check_expr('5.4(i)', q5_4a, '11938aebfa47')
verify_sketch('5.4(ii)', q5_4b, x**2/2 + 5*x + 13)
```

6 · Sketch a curve from its formula, labelling what was asked for

The largest section: **11 questions, 33 marks**. Nine of them are on calculator papers, and here the calculator earns its place — nobody finds the roots of $e^x - 3x - 4$ by hand for a sketch.

Read the instruction first and write down the list of things it names. The marks are for the labels.

6.1 · May 2021 TZ1, Paper 3, Q1(a)(i) · 3 marks · calculator

This question asks you to explore the behaviour and key features of cubic polynomials of the form $x^3 - 3cx + d$.

Consider $f(x) = x^3 - 3cx + 2$ for $x \in \mathbb{R}$, where c is a parameter.

Sketch the graph of $y = f(x)$ showing the value of the y -intercept and the coordinates of any points with zero gradient, for $c = 1$.

```
q6_1 = {...}      # keys: y_intercept, maxima,
minima

f61 = x**3 - 3*x + 2
verify_sketch('6.1', q6_1, f61)

show(f61, span=(-3, 3), ylim=(-6, 10))
```

6.2 · May 2021 TZ1, Paper 3, Q1(a) (ii) · 3 marks · calculator

The same, for $c = 2$: sketch $y = x^3 - 6x + 2$, showing the value of the y -intercept and the coordinates of any points with zero gradient.

The markscheme accepts the exact coordinates as well as the 3 s.f. ones.

```
q6_2 = {...}      # keys: y_intercept, maxima,
minima

f62 = x**3 - 6*x + 2
verify_sketch('6.2', q6_2, f62)

show(f62, span=(-3, 3), ylim=(-8, 12))
```

6.3 · May 2022 TZ1, Paper 2, Q10(a) · 2 marks · calculator

Consider the function $f(x) = \sqrt{x^2 - 1}$, where $1 \leq x \leq 2$.

Sketch the curve $y = f(x)$, clearly indicating the coordinates of the endpoints.

```
q6_3 = {...}      # key: endpoints

f63 = sqrt(x**2 - 1)
verify_sketch('6.3', q6_3, f63, domain=(1,
2))

show(f63, span=(1, 2), size=(5, 3))
```

6.4 · May 2022 TZ1, Paper 3, Q2(c) · 2 marks · calculator

Consider $f(x) = (x - 1)(x^2 - 8x + 17)$ for $x \in \mathbb{R}$. Part (b) has shown that the line $y = x - 1$ is tangent to the curve $y = f(x)$ at the point A(4, 3).

Sketch the curve $y = f(x)$ and the tangent to the curve at point A, clearly showing where the tangent crosses the x -axis.

```

q6_4 = {...}      # key: x_intercepts – for
the curve
q6_4b = {...}     # key: x_intercepts – for
the tangent line

f64 = (x - 1)*(x**2 - 8*x + 17)
verify_sketch('6.4 curve', q6_4, f64)
verify_sketch('6.4 tangent', q6_4b, x - 1)

show(f64, (x - 1, 'tangent at A'), span=(-1,
6), ylim=(-10, 25),
marks=[(4, 3)])

```

6.5 · May 2022 TZ1, Paper 3, Q2(h)(i)

· 2 marks · calculator

Consider the curve $y = (x - r)(x^2 - 2ax + a^2 + b^2)$ in the special case $a = r = 1$ and $b = 2$.

Sketch the curve.

*The markscheme wants a positive cubic with **no** stationary points and a non-stationary point of inflexion at $x = 1$. State the intercepts and the turning points you find — and there is a right answer for a list that turns out to be empty.*

```
q6_5 = {...}      # keys: x_intercepts,
maxima, minima

f65 = (x - 1)*(x**2 - 2*x + 5)
verify_sketch('6.5', q6_5, f65)

show(f65, span=(-2, 4), ylim=(-20, 20))
```

6.6 · November 2022, Paper 1, Q11(b)(iv) · 3 marks · no calculator

Let $P(x) = x^3 + ax^2 + bx + c$, where each of a, b, c is one of 1, 2, 3, 4, 5 and no value is repeated. Earlier parts give $a = 4, b = 5, c = 2$, so $P(x) = (x + 1)^2(x + 2)$.

Hence or otherwise, sketch the graph of $y = P(x)$, clearly showing the coordinates of any intercepts with the axes.

```
q6_6 = {...}      # keys: x_intercepts,
y_intercept, maxima, minima

f66 = x**3 + 4*x**2 + 5*x + 2
verify_sketch('6.6', q6_6, f66)

show(f66, span=(-3, 1), ylim=(-3, 6))
```


6.7 · May 2023 TZ1, Paper 3, Q1(a) · 4 marks · calculator

In this question you will be investigating the family of functions of the form $f(x) = x^n e^{-x}$.

When $n = 1$, the function is $f_1(x) = x e^{-x}$, where $x \geq 0$.

Sketch the graph of $y = f_1(x)$, stating the coordinates of the local maximum point.

```
q6_7 = {...}      # keys: x_intercepts, maxima

f67 = x*exp(-x)
verify_sketch('6.7', q6_7, f67, domain=(0,
oo))

show(f67, span=(0, 8), size=(5.5, 3))
```

6.8 · November 2023, Paper 2, Q2(a) · 3 marks · calculator

Recorded twice in the corpus, TZ1 and TZ2. One paper.

Consider the function $f(x) = e^x - 3x - 4$.

On the axes provided, sketch the graph of f for $-4 \leq x \leq 3$.

The markscheme pays for the roots, for the y -intercept together with the local minimum, and for the two endpoints.

```
q6_8 = {...}      # keys: x_intercepts,
y_intercept, minima, endpoints

f68 = exp(x) - 3*x - 4
verify_sketch('6.8', q6_8, f68, domain=(-4,
3))

show(f68, span=(-4, 3), ylim=(-6, 10))
```

6.9 · May 2025 TZ1, Paper 1, Q10(b) · 4 marks · no calculator

The function f is defined by $f(x) = 5(x + 1)(x + 3)$, where $x \in \mathbb{R}$.

Sketch the graph of $y = f(x)$, showing the values of any intercepts with the axes and the coordinates of the vertex.

```
q6_9 = {...}      # keys: x_intercepts,
y_intercept, minima

f69 = 5*(x + 1)*(x + 3)
verify_sketch('6.9', q6_9, f69)

show(f69, span=(-5, 1), ylim=(-8, 20))
```

6.10 · May 2025 TZ3, Paper 2, Q12(c)
· 5 marks · calculator

Part (b) has solved $\frac{dy}{dx} = x^2 - y - 5$ and given the general solution $y = x^2 - 2x - 3 + Ce^{-x}$.

Sketch the curve of the particular solution which passes through the point $(-3, 2)$, for $-4 \leq x \leq 4$, clearly labelling the coordinates of any local maximum and minimum points.

```
q6_10 = {...}      # keys: maxima, minima

f610 = x**2 - 2*x - 3 - 10*exp(-x - 3)
verify_sketch('6.10', q6_10, f610)

show(f610, span=(-4, 4), ylim=(-6, 6))
```

6.11 · November 2025 TZ1, Paper 3, Q1(d)(ii) · 2 marks · calculator

This question asks you to investigate lines normal to curves of the form $y = \frac{k^2}{x}$.

The curve H is $y = \frac{1}{x}$. Earlier parts have shown that a line normal to H at $x = t$ has gradient t^2 , and that there are

exactly two normals of gradient m whenever $m > 0$, $m \neq 1$.

For an appropriate value of m , sketch two lines that are normal to H , clearly indicating the point at which each line is normal to H .

Take $m = 4$ and give the two values of x at which a line of gradient 4 is normal to H .

```
q6_11 = [...]                # the two
values of x

# the normal's gradient is -1 over the
curve's gradient
verify_roots('6.11', q6_11, -1/diff(1/x, x) -
4, (-10, 10))

show(1/x, (4*x + 1/2 - 8, 'normal at x = 2'),
(4*x - 1/2 + 8, 'normal at x = -2'),
      span=(-4, 4), ylim=(-6, 6), marks=[(2,
Rational(1, 2)), (-2, Rational(-1, 2))])
```

7 · Sketch using the second derivative

1 question, 3 marks — and it is the only place in the topic where the mark is

for the shape *between* the features, which no check can see. So this one is answered through the second derivative instead: the numbers that decide how sharply the curve turns.

**7.1 · November 2023, Paper 1,
Q11(c) · 3 marks · no calculator**

Recorded twice in the corpus, TZ1 and TZ2. One paper.

Consider $f(x) = e^{\cos 2x}$, where $-\frac{\pi}{4} \leq x \leq \frac{5\pi}{4}$. Part (a) found the points of zero gradient and part (b) used the second derivative to classify them.

Sketch the curve of $y = f(x)$ for $0 \leq x \leq \pi$, taking into consideration the relative values of the second derivative found in part (b).

(i) $f''(0)$. **(ii)** $f''(\frac{\pi}{2})$. **(iii)** By what factor is the curve's bend sharper at the maximum than at the minimum, that is $|f''(0)| \div |f''(\pi/2)|$?

```

q7_1a = ...                # f''(0),
exactly
q7_1b = ...                # f''(pi/2),
exactly
q7_1c = ...                # the ratio of
the magnitudes

check_expr('7.1(i)', q7_1a, '2e78d6cb355c')
check_expr('7.1(ii)', q7_1b, '281e7b9cac6c')
check_expr('7.1(iii)', q7_1c, '31931e836b92')

show(exp(cos(2*x)), span=(0, float(pi)),
size=(5.5, 3),
      marks=[(0, float(E)), (float(pi/2),
float(1/E)), (float(pi), float(E))])

```

8 · Sketch a curve that is not a function

$y^2 = f(x)$: two branches, mirror images in the x -axis, and at an x -intercept the curve is vertical — unless the two branches meet there in a cusp. **5 questions, 10 marks, all of them Paper 3.**

8.1 · May 2022 TZ2, Paper 3, Q1(a)(i)
· 2 marks · calculator

This question asks you to explore properties of a family of curves of the type $y^2 = x^3 + ax + b$.

On the same set of axes, sketch $y^2 = x^3$ for $x \geq 0$, on $-2 \leq x \leq 2$ and $-2 \leq y \leq 2$, clearly indicating any points of intersection with the coordinate axes.

The upper branch is $y = x^{3/2}$. Give its intercepts, and its gradient at the origin — that number is what makes the markscheme's cusp a cusp.

```
q8_1 = {...}      # keys: x_intercepts,
y_intercept - for the upper branch
q8_lb = ...       # the gradient of the upper
branch at the origin

verify_sketch('8.1', q8_1, x**Rational(3, 2),
domain=(0, 2))
check_num('8.1 gradient', q8_lb, 6,
'5feceb66ffc8')

show(x**Rational(3, 2), -x**Rational(3, 2),
span=(0, 2), ylim=(-2, 2), size=(5, 3.2))
```

8.2 · May 2022 TZ2, Paper 3, Q1(a)

(ii) · 2 marks · calculator

On the same axes, sketch $y^2 = x^3 + 1$ for $x \geq -1$, clearly indicating any points of intersection with the coordinate axes.

```
q8_2 = {...}      # key: x_intercepts – for
the upper branch
q8_2b = [...]     # both y-intercepts of the
curve

verify_sketch('8.2', q8_2, sqrt(x**3 + 1),
domain=(-1, 2))
check_set('8.2 y-intercepts', q8_2b,
'c716ad9a564d')

show(sqrt(x**3 + 1), -sqrt(x**3 + 1), span=
(-1, 2), ylim=(-3, 3), size=(5, 3.2))
```

8.3 · May 2022 TZ2, Paper 3, Q1(b)

(ii) · 1 mark · calculator

By considering each curve from part (a), identify **two** key features that would distinguish one curve from the other.

The exam wants words, and the solution lists every accepted answer. One of them is a number, so answer that one here: the gradient of the upper branch of $y^2 = x^3 + 1$ at its x -intercept.


```
q8_3 = ... # a number, or
00
check_expr('8.3', q8_3, 'a8c23cc81417')
```

8.4 · May 2022 TZ2, Paper 3, Q1(c) · 2 marks · calculator

Now consider curves of the form $y^2 = x^3 + b$, for $x \geq -\sqrt[3]{b}$, where $b \in \mathbb{Z}^+$.

By varying the value of b , suggest **two** key features common to these curves.

Words again, and no check: this one is scored by the solution below. Write your two down first, then compare — the markscheme prints eleven acceptable answers and refuses one that looks acceptable.

8.5 · November 2023, Paper 3, Q2(b) · 3 marks · calculator

Recorded twice in the corpus, TZ1 and TZ2. One paper.

A family of curves has equation $y^2 = 4a^2 - 4ax$, and a second family has

equation $y^2 = 4b^2 + 4bx$, where a and b are positive real parameters.

Consider the case $a = 2$ and $b = 1$. On the same set of axes, sketch the curves $y^2 = 16 - 8x$ and $y^2 = 4 + 4x$. On your sketch, clearly label each curve and any x -intercepts.

```
q8_5a = {...}      # key: x_intercepts - for
y^2 = 16 - 8x
q8_5b = {...}      # key: x_intercepts - for
y^2 = 4 + 4x

verify_sketch('8.5 first', q8_5a, sqrt(16 -
8*x), domain=(-5, 2))
verify_sketch('8.5 second', q8_5b, sqrt(4 +
4*x), domain=(-1, 5))

show((sqrt(16 - 8*x), 'y^2 = 16 - 8x'), (-
sqrt(16 - 8*x), None),
      (sqrt(4 + 4*x), 'y^2 = 4 + 4x'), (-
sqrt(4 + 4*x), None),
      span=(-2, 3), ylim=(-6, 6))
```

9 • Count the roots of a family

A letter in the formula and a question that asks for a count. The count changes exactly when a turning point crosses the

axis — which is why this is the last section: it needs the whole list and one idea more.

9 questions, 14 marks, and all of them are one Paper 3 investigation:

November 2023, the family $y = x^3 + ax^2 + b$.

The check here holds no answer at all. It is handed a function that counts the distinct real roots of the cubic, and it asks that function about your set — inside, outside, and on both sides of every endpoint.

```
# The whole of section 9 is checked against
this. It is not an answer: it
# is the question, written so a machine can
be asked it one b at a time.
def crossings(a_value, count):
    def holds(b_value):
        roots = Poly(x**3 + a_value*x**2 +
b_value, x).real_roots()
        return len(set(roots)) == count
    return holds

print('ready')
```

9.1 · November 2023, Paper 3, Q1(a)
· 2 marks · calculator

This question asks you to explore some properties of the family of curves $y = x^3 + ax^2 + b$, where a, b are real parameters.

First consider the case $a = 3$.

By systematically varying the value of b , or otherwise, find the two values of b such that the curve $y = x^3 + 3x^2 + b$ has exactly two x -axis intercepts.

```
q9_1 = ... # a set:
FiniteSet(...)

verify_param_set('9.1', q9_1, crossings(3,
2), var=b)
```

9.2 · November 2023, Paper 3, Q1(b)
(i) · 1 mark · calculator

Write down the set of values of b such that $y = x^3 + 3x^2 + b$ has exactly **one** x -axis intercept.

```
q9_2 = ... #  
Interval.open(...), Union(...), ...  
  
verify_param_set('9.2', q9_2, crossings(3,  
1), var=b)
```

9.3 · November 2023, Paper 3, Q1(b) (ii) · 1 mark · calculator

Write down the set of values of b such that $y = x^3 + 3x^2 + b$ has exactly **three** x -axis intercepts.

```
q9_3 = ...  
  
verify_param_set('9.3', q9_3, crossings(3,  
3), var=b)
```

9.4 · November 2023, Paper 3, Q1(c) (i) · 1 mark · calculator

Now consider the case $a = -3$. Write down the set of values of b such that $y = x^3 - 3x^2 + b$ has exactly **two** x -axis intercepts.

```
q9_4 = ...  
  
verify_param_set('9.4', q9_4, crossings(-3,  
2), var=b)
```

9.5 · November 2023, Paper 3, Q1(c)
(ii) · 1 mark · calculator

Write down the set of values of b such that $y = x^3 - 3x^2 + b$ has exactly **one** x -axis intercept.

```
q9_5 = ...
```

```
verify_param_set('9.5', q9_5, crossings(-3,  
1), var=b)
```

9.6 · November 2023, Paper 3, Q1(c)
(iii) · 1 mark · calculator

Write down the set of values of b such that $y = x^3 - 3x^2 + b$ has exactly **three** x -axis intercepts.

```
q9_6 = ...
```

```
verify_param_set('9.6', q9_6, crossings(-3,  
3), var=b)
```

9.7 · November 2023, Paper 3, Q1(d)
· 1 mark · calculator

Consider the curve $y = x^3 + ax^2 + b$ for $a \neq 0$, in the case where it has exactly three x -axis intercepts.

State whether each point of zero gradient is located above or below the x -axis.

One line of words, and no check — this is the sentence part (h) is built on. Write it down before opening the solution.

**9.8 · November 2023, Paper 3, Q1(f)
(ii) · 1 mark · calculator**

Consider the points $P(0, b)$ and $Q(-\frac{2}{3}a, \frac{4}{27}a^3 + b)$ of zero gradient for $a > 0$ and $b > 0$.

Determine whether each point is located above or below the x -axis.

Words again, no check.

**9.9 · November 2023, Paper 3, Q1(h)
· 5 marks · calculator**

Prove that if $4a^3b + 27b^2 < 0$ then the curve $y = x^3 + ax^2 + b$ has exactly three x -axis intercepts.

The last five marks of the investigation, and a proof — nothing to check. Write it

out, then read the solution: the whole argument is 9.7 said in algebra.

Solutions

Numbered exactly as the questions. Each one gives the answer, the shortest route to it, and the markscheme detail — which is usually where the mark was actually lost.

1 • Reading them off

1.1 $x = 3$ and $y = -2$. The denominator $3 - x$ vanishes at $x = 3$; the degrees are equal so the horizontal asymptote is the ratio of the leading coefficients, $\frac{2}{-1} = -2$.

That minus sign is the whole difficulty. Written as $\frac{2x + 4}{3 - x}$ the denominator's leading coefficient is -1 , and reading the ratio as 2 is the standard slip; rewriting as $\frac{-2x - 4}{x - 3}$ first makes it visible.

1.2 $x = \frac{15}{2}$. The markscheme prints "Award A0 for $x \neq \frac{15}{2}$ " — copying the exclusion from the stem is not an answer. It also allows the mark to be earned in

part (d) if the line appears on the sketch there.

1.3 $x = -1, y = 2$.

1.4 $x = 2, y = \frac{7}{2}$. The markscheme writes *"must be an equation with x "* and *"must be an equation with y "* against the two A1s.

1.5 $x = 2, y = 1$. Here the function is already in the shifted form $1 - \frac{1}{x-2}$, so both asymptotes can be read straight off: the subtracted fraction blows up at $x = 2$ and dies at infinity, leaving $y = 1$.

1.6 $x = -3$. Accept $2x + 6 = 0$.

1.7 $y = \frac{2}{3}$. Cancelling the constants first — $\frac{2(x+3)}{3(x+2)}$ — hides nothing: the leading coefficients are 2 and 3.

1.8 $y = \frac{3}{2}$. *"Must be an equation. Do not accept the $\neq$ sign."*

2 • The ones you cannot read off

2.1 $y = \frac{\pi}{2}$. As $x \rightarrow \pm\infty$ the inside tends to 1, so $f(x) \rightarrow \arcsin 1 = \frac{\pi}{2}$. Two marks: one for the limit argument, one for the equation.

The trap is stopping at the inside. $\frac{x^2 - 1}{x^2 + 1} \rightarrow 1$ is true and worth nothing on its own — the asymptote is $\arcsin$ of that, not that.

2.2 $x = 0$. Nothing is a denominator here. The graph of $g(x) = 2 \ln x - \ln d$ stops where its domain stops, and $\ln x \rightarrow -\infty$ as $x \rightarrow 0^+$ whatever d is. The value of d never enters — which is exactly what the single mark is testing.

3 • The oblique one

3.1 $x = -\frac{2}{3}$ and $y = \frac{4}{3}x - \frac{8}{9}$, and **there is no horizontal asymptote**. The numerator's degree is one higher than the denominator's, so the ratio of the leading coefficients, $\frac{4}{3}$, is the *gradient* of

a slanted line, not a horizontal one. Quoting $y = \frac{4}{3}$ here is the single most common way to lose the part.

The markscheme splits its four marks A1 (the vertical), A1 (the gradient $\frac{4}{3}$), M1 (a method for the whole line), A1 (the equation) — and then notes *"Do not award the final A1 if the answer is not given as an equation."* Long division:

$$\frac{4x^2 - 1}{3x + 2} = \frac{4}{3}x - \frac{8}{9} + \frac{7/9}{3x + 2}.$$

3.2 $y = \frac{x}{2} + \frac{13}{4}$, so $a = \frac{1}{2}$, $b = \frac{13}{4}$. Three routes, all in the markscheme: divide; or write $\frac{x^2 - x - 12}{2x - 15} \equiv \frac{x}{2} + b + \frac{c}{2x - 15}$ and equate coefficients of x , giving $-1 = -\frac{15}{2} + 2b$; or take $a = \lim_{x \rightarrow \infty} \frac{f(x)}{x}$ and $b = \lim_{x \rightarrow \infty} (f(x) - ax)$.

The sign in $-1 = -\frac{15}{2} + 2b$ is where the part is lost: dropping the minus gives $b = -\frac{13}{4}$.

3.3 $y = \frac{x}{2} - \frac{17}{2}$, so $a = \frac{1}{2}$, $b = -\frac{17}{2}$. Same three routes; equating coefficients of x

gives $-14 = 3 + 2b$.

3.4 $y = x$, for **every** positive A . Part (d)(i) has already done the division for you:

$$\frac{x(x^2 - A)}{x^2 + A} = x - \frac{2Ax}{x^2 + A}, \quad \text{and} \quad \text{the}$$

subtracted term tends to 0 as $x \rightarrow \pm\infty$ because its denominator has the higher degree. *"Award A0 if not given as an equation."*

4 • Rational sketches

4.1 Vertical asymptotes $x = \pm\frac{1}{2}$, horizontal asymptote $y = 0$, x -intercept $-\frac{2}{3}$, y -intercept -2 , local minimum $(-1.11, -0.339)$ on the left branch and local maximum $(-0.226, -1.66)$ on the middle one. Three branches, not two.

The markscheme is worth reading in full: A1 for both vertical asymptotes with their equations, A1 for the horizontal asymptote with its equation, A1 for each correct branch including asymptotic behaviour, the coordinates of the minimum and maximum, and the values

of the axes intercepts — and then *"If vertical asymptotes are absent (or not vertical) and the branches overlap as a consequence, award maximum A0A1A0A1A1."* Overlapping branches are the failure this question is built to catch.

4.2 $x = 3$, $y = -2$, $(-2, 0)$, $(0, \frac{4}{3})$. One mark, for *"completely correct shape: two branches in correct quadrants with asymptotic behaviour"*.

4.3 $x = \frac{15}{2}$, $y = \frac{x}{2} + \frac{13}{4}$, $(-3, 0)$, $(4, 0)$, $(0, \frac{4}{5})$. Three marks: shape, both asymptotes with correct asymptotic behaviour, intercepts — and the markscheme adds *"Points of intersection with the axes and the equations of asymptotes are not required to be labelled."* Here the drawing itself is the answer.

4.4 $x = -1$, $y = 2$, $(\frac{1}{2}, 0)$, $(0, -1)$. Three marks: one for the shape with both asymptotes, one for each intercept.

4.5 Vertical asymptotes $x = -1$ and $x = 3$, horizontal asymptote $y = 0$, **no** x -

intercept at all, y -intercept $-\frac{1}{3}$, local maximum $(1, -\frac{1}{4})$.

Two vertical asymptotes cut the line into three pieces, so the curve has three branches, and the middle one is the only one carrying a turning point. The maximum sits at $x = 1$ because that is the axis of symmetry of $x^2 - 2x - 3$ — which is a legitimate method here and a wrong habit generally; the markscheme allows *"uses the axis of symmetry or attempts to solve $f'(x) = 0$ "*. It then awards (M1)A0 if the maximum is placed at $x = 1$ without its coordinates.

4.6 $x = 2$, $y = 1$, $(3, 0)$, $(0, \frac{3}{2})$.

4.7 $x = -3$, $y = \frac{x}{2} - \frac{17}{2}$, $(2, 0)$, $(12, 0)$, $(0, 4)$. Four marks, and the first has a condition attached: *"two branches with approximately correct shape — for this A1 the graph must be a function."* A curve drawn as a loop earns nothing for shape.

4.8 Crossings at $(-4, 0)$, $(0, 0)$, $(4, 0)$; local maximum $(-1.94, 1.20)$, local minimum

(1.94, -1.20). The curve is odd, which is why the two turning points are reflections of each other through the origin.

4.9 $(-4, 0)$, $(0, 0)$, $(4, 0)$ — and the markscheme adds "*Award AO if additional points are given.*" The factor x in front is easy to drop, and dropping it loses the intercept at the origin.

4.10 $x = 0$ and $y = 0$; two branches, in the first and third quadrants. One mark, and it is the setup for the two normals in 6.11.

5 • Ranges

5.1 $y \geq -5$. Complete the square: $f(x) = 6(x - 1)^2 - 5$, so the vertex is $(1, -5)$ and the parabola opens upwards. The vertex is *attained*, so the inequality is not strict.

5.2 $g(x) \leq \frac{5 - \sqrt{13}}{6}$ or $g(x) \geq \frac{5 + \sqrt{13}}{6}$, that is $g(x) \leq 0.232$ or $g(x) \geq 1.43$.

Set $y = \frac{2x-5}{x^2-3}$ and clear the fraction:
 $yx^2 - 2x + (5 - 3y) = 0$. For a real x to exist the discriminant must be non-negative:
 $4 - 4y(5 - 3y) = 12y^2 - 20y + 4 \geq 0$, whose roots are $\frac{5 \pm \sqrt{13}}{6}$. Both ends are attained — they are the turning points of the curve — so both inequalities are weak, and there is a gap between them because the two branches never meet.

The calculator route (find the local maximum and minimum) gets the same answer to 3 s.f. and cannot produce the surd. Both are accepted.

5.3 $y \leq -10 - 5\sqrt{3}$ or $y \geq -10 + 5\sqrt{3}$, that is $y \leq -18.7$ or $y \geq -1.34$.

The markscheme: *A1* for each inequality, and then *"Award A1A0 for strict inequalities in both."* Both ends are turning points of the curve, both are reached, and writing $<$ costs a mark even though the two numbers are right. This is the whole reason `verify_range` tests each endpoint separately.

5.4 Axis of symmetry $x = -5$; minimum point $(-5, \frac{1}{2})$.

With $k = 5$ the function is $\frac{1}{2}x^2 + 5x + 13$, so the axis is $x = -\frac{b}{2a} = -\frac{5}{2 \cdot \frac{1}{2}} = -5$; substituting back gives $\frac{1}{2}(25) - 25 + 13 = \frac{1}{2}$. Completing the square does both at once: $\frac{1}{2}(x + 5)^2 + \frac{1}{2}$. *"Must be an equation"* against the axis.

6 • Labelled sketches

6.1 y -intercept 2; local maximum $(-1, 4)$; local minimum $(1, 0)$. $f'(x) = 3x^2 - 3c$ with $c = 1$ gives $x = \pm 1$, and $f(1) = 0$ means the curve *touches* the axis there rather than crossing it: $x^3 - 3x + 2 = (x - 1)^2(x + 2)$.

6.2 y -intercept 2; local maximum $(-\sqrt{2}, 2 + 4\sqrt{2}) = (-1.41, 7.66)$; local minimum $(\sqrt{2}, 2 - 4\sqrt{2}) = (1.41, -3.66)$. The markscheme prints the exact coordinates as an accepted alternative — worth knowing, because the calculator will not offer them.

6.3 Endpoints $(1, 0)$ and $(2, \sqrt{3}) = (2, 1.73)$, and the shape is concave down. Two marks: one for the shape on the given domain, one for both endpoints. *"The coordinates of endpoints may be seen on the graph or marked on the axes."*

The domain is given as $1 \leq x \leq 2$ and the curve simply stops there. Drawing the whole of $\sqrt{x^2 - 1}$ loses the shape mark.

6.4 The curve crosses the x -axis at $x = 1$ only — its quadratic factor $x^2 - 8x + 17 = (x - 4)^2 + 1$ has no real roots — and so does the tangent $y = x - 1$. That coincidence is what the question is about: *"clearly showing where the tangent crosses the x -axis"*, and the markscheme adds *"Award A1A0 if both graphs cross the x -axis at distinctly different points."*

The other mark is for a positive cubic with a local maximum and local minimum in the first quadrant, both to the left of A(4, 3). The minimum is at $x = \frac{10 + \sqrt{7}}{3} \approx 4.2$ — so close to A that the markscheme

condones sketches where they appear to coincide.

6.5 x -intercept 1, and **no stationary points at all**: the derivative $3x^2 - 6x + 7$ has discriminant $36 - 84 < 0$. The curve rises through a non-stationary point of inflexion at $x = 1$.

Drawing a cubic with two turning points because cubics have two is the trap. The markscheme: *"a positive cubic with no stationary points and a non-stationary point of inflexion at $x = 1$ ", with the note "graphs may appear approximately linear; award this A1 if a change of concavity either side of $x = 1$ is apparent."*

6.6 x -intercepts -2 and -1 , y -intercept 2, local maximum $(-\frac{5}{3}, \frac{4}{27})$, local minimum $(-1, 0)$.

$P(x) = (x + 1)^2(x + 2)$, so $x = -1$ is a double root: the curve *touches* the axis there, and that touch is the local minimum. The markscheme asks for the local maximum only to be *"anywhere between $x = -2$ and*

$x = -1$ " but insists on the local minimum being at $(-1, 0)$.

6.7 Passes through the origin; local maximum $(1, \frac{1}{e}) = (1, 0.368)$; domain $x \geq 0$; single maximum with $y \rightarrow 0$ as $x \rightarrow \infty$.

Four separate A1s, and one of them is *for the correct domain* — the family is defined for $x \geq 0$ and drawing the left half throws a mark away. The asymptote does not have to be named.

6.8 Roots -1.24 and 2.42 ; y -intercept -3 ; local minimum $(\ln 3, -3 \ln 3 - 1) = (1.10, -4.30)$; endpoints $(-4, 8.02)$ and $(3, 7.09)$.

Three marks, and the third is *for the endpoints* — with the intervals spelled out: the left end in $-4.5 < x < -3.5$, $7.5 < y < 8.5$ and the right end in $2.5 < x < 3.5$, $6.5 < y < 7.5$. A sketch that runs off the given window is a different sketch.

6.9 x -intercepts -3 and -1 , y -intercept 15 , vertex $(-2, -5)$. Part (a) has already written f as $5(x + 2)^2 - 5$, which hands

you the vertex; the intercepts are the factors and $f(0) = 15$. Four marks: shape, roots, y -intercept, vertex.

6.10 Local maximum $(-2.70, 2.28)$, local minimum $(0.899, -4.19)$, three x -intercepts — two negative and one positive — and the curve is drawn only on $-4 \leq x \leq 4$.

The particular solution comes first: $2 = 9 + 6 - 3 + Ce^3$ gives $C = -10e^{-3} = -0.498$, so $y = x^2 - 2x - 3 - 10e^{-x-3}$. Five marks, of which one is *for the domain* and one is for the two labelled turning points together.

6.11 $x = 2$ and $x = -2$: the normal at $x = t$ has gradient t^2 , so gradient 4 means $t^2 = 4$. The two normals are $y = 4x - \frac{15}{2}$ and $y = 4x + \frac{15}{2}$, touching H at $(2, \frac{1}{2})$ and $(-2, -\frac{1}{2})$.

The markscheme wants two *parallel* lines, each normal to H , each with $c \neq 0$, and it adds: "*for normal lines that cross the y -axis, award A1A0 if the y -intercepts are not approximately equidistant from O*", and "*maximum A1A0 for a normal line*

sketched for $0 < m < 1$ and a normal line sketched for $m > 1$ " — that is, the two lines must share one m , which is the entire point of part (c)(ii).

7 • Shape from the second derivative

7.1 $f''(0) = -4e \approx -10.9$, $f''(\frac{\pi}{2}) = \frac{4}{e} \approx 1.47$, and the ratio of the magnitudes is $e^2 \approx 7.39$.

Differentiate twice: $f'(x) = -2 \sin(2x) e^{\cos 2x}$ and

$$f''(x) = (4 \sin^2(2x) - 4 \cos 2x) e^{\cos 2x}.$$

At $x = 0$ the sine vanishes and $\cos 0 = 1$, leaving $-4e$; at $x = \frac{\pi}{2}$ the sine vanishes again and $\cos \pi = -1$, leaving $4e^{-1}$.

The sketch: maxima at $(0, e)$ and (π, e) , minimum at $(\frac{\pi}{2}, \frac{1}{e})$ — and the third mark is *"for showing a higher rate of change of gradient at the maxima and a lower rate of change of gradient at the minimum point."* Seven times sharper is what those two numbers mean: two narrow

peaks and a long flat trough. The word *relative* in the question is the whole question.

8 · Curves that are not functions

8.1 The curve passes through the origin and meets the axes nowhere else; the gradient of the upper branch $y = x^{3/2}$ at the origin is 0, and the lower branch arrives with gradient 0 too, from below. Two zero gradients meeting head-on is a **cusp** — and the markscheme's two marks are exactly "*approximately symmetric about the x-axis graph of $y^2 = x^3$* " and "*including cusp/sharp point at (0,0).*"

8.2 x -intercept -1 , y -intercepts 1 and -1 . Marks for the symmetric shape "*with approximately correct gradient at axes intercepts*" and for the positions $x = -1$, $y = \pm 1$.

The two "gradients at the intercepts" are the whole distinction from 8.1, which is

what the next part asks for.

8.3 ∞ — the branch is vertical there. Differentiating $y^2 = x^3 + 1$ implicitly, $2y y' = 3x^2$, and at $(-1, 0)$ the left side vanishes while the right does not, so y' is undefined and the tangent is vertical.

The question wants words, and the markscheme accepts any **two** of: $y^2 = x^3$ has a cusp and the other does not; the graphs have different domains; $y^2 = x^3 + 1$ has points of inflexion and the other does not; different x -intercepts (one goes through the origin, the other does not); different y -intercepts.

8.4 Any **two** of the eleven the markscheme lists: as $x \rightarrow \infty$, $y \rightarrow \pm\infty$; for large x the curve is approximated by $y^2 = x^3$; one x -intercept, at $x = -\sqrt[3]{b}$; two y -intercepts, at $y = \pm\sqrt{b}$; they all have the same range; $y = 0$ is a line of symmetry; one x -intercept; two y -intercepts; two points of inflexion; vertical gradient at the x -intercept; no cusp at the x -intercept.

And the refusal: *"Do not credit an answer of 'they are all symmetrical' without some reference to the line of symmetry", and "do not allow same/similar shape".* A key feature has to be a feature, not an impression.

8.5 $y^2 = 16 - 8x$ opens to the **left** from its x -intercept 2; $y^2 = 4 + 4x$ opens to the **right** from -1 . They cross twice, once in the first quadrant and once in the fourth. Three marks: one for each curve having approximately the right shape and position, one for the two x -intercepts.

The general result waiting two parts later is that these two families cross at right angles — which is why the sketch is asked for first.

9 • Counting the roots of $y = x^3 + ax^2 + b$

The one idea behind all nine parts: the cubic has zero gradient at $P(0, b)$ and at $Q(-\frac{2}{3}a, \frac{4}{27}a^3 + b)$, and it has

- **three** x -intercepts when P and Q are on opposite sides of the axis,
- **two** when one of them is *on* the axis,
- **one** when both are on the same side.

Everything below is that sentence with numbers in it.

9.1 $b = -4$ and $b = 0$. With $a = 3$ the two zero-gradient points are $(0, b)$ and $(-2, 4 + b)$; they lie on the axis when $b = 0$ and when $b = -4$. Two marks, one of them (M1) for "*varies the value of b with $a = 3$* " or for evidence that the case $b = 0$ was considered at all.

9.2 $b < -4$ or $b > 0$. **9.3** $-4 < b < 0$.

9.4 $b = 0$ and $b = 4$. With $a = -3$ the points are $(0, b)$ and $(2, b - 4)$.

9.5 $b < 0$ or $b > 4$. **9.6** $0 < b < 4$.

The pair (9.2, 9.3) and the pair (9.5, 9.6) are the same two answers with the interval reflected, because changing the sign of a reflects the family in the y -axis.

9.7 One point of zero gradient is above the x -axis and the other is below it — one on either side. That is the whole mark.

9.8 Both above. With $a > 0$ and $b > 0$: $P = (0, b)$ has $b > 0$, and Q has height $\frac{4}{27}a^3 + b$, a sum of two positive numbers. (Part (f)(i) has just shown $\frac{d^2y}{dx^2} = 6x + 2a$, so P is the local minimum and Q the local maximum — and if even the minimum is above the axis, the curve crosses once.)

9.9 Factorise the given quantity:

$$4a^3b + 27b^2 = 27b \left(\frac{4}{27}a^3 + b \right) < 0.$$

The two factors are exactly the heights of P and Q — b and $\frac{4}{27}a^3 + b$ — up to the positive constant 27. Their product is negative precisely when exactly one of the two heights is negative, that is when P and Q lie on opposite sides of the x -axis; and by 9.7 that happens if and only if the curve has exactly three x -axis intercepts.

Five marks: M1 for attempting the factorisation, A1 for it, A1 for stating *both* cases ($b > 0$ with $\frac{4}{27}a^3 + b < 0$, or $b < 0$ with $\frac{4}{27}a^3 + b > 0$ — "*only award this A1 if both cases are stated*"), R1 for opposite signs meaning opposite sides, R1 for opposite sides meaning three intercepts. Proving the converse instead earns at most three of the five.

Done. Fifty-four questions, 127 marks, the whole of the topic as the archive has it.

What the checks could not see, and you should: the shape of every curve *between* the features you listed, which is where the sketching marks divide; the two verbal parts, 8.4 and 9.7; and the proof in 9.9. For those the solution above is the marker.

If a section came out slow, that section's rung in *Practicum B4* has the theory, the traps, and the drill items behind it.